

A Project report on

DRIVER DROWSINESS DETECTION

*Submitted in partial fulfillment of the requirements
for the award of the degree of*

BACHELOR OF TECHNOLOGY

in

Computer Science & Engineering

By

N.PRAGATHI

164G1A0568

K.PRANEETHA

164G1A0569

B.RAVEENA SAI

164G1A0578

Y. SREEDHAR

JASHWANTH

164G1A05A1

S.VIJAYKUMAR

164G1A05B7

Under the Guidance of

Mr. M. Narasimhulu, M.Tech, (PhD)

Assistant Professor



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(B. Tech program accredited by NBA)

SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY: ANANTAPURAMU

(Accredited By NAAC With 'A' Grade, Affiliated To JNTUA, Approved By AICTE, New Delhi)

2019-2020

SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY: ANANTHAPURAMU

(Accredited by NAAC with 'A' Grade, Affiliated to JNTUA, Approved by AICTE, New Delhi) Rotarypuram Village, B.K. Samudram Mandal, Ananthapuramu – 515701

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(B. Tech Program accredited by NBA)



Certificate

This is to certify that the project report entitled **DRIVER DROWSINESS SYSTEM** is the bonafide work carried out by **N. Pragathi** bearing Roll Number **164G1A0568**, **K. Praneetha** bearing Roll Number **164G1A0569**, **B. Raveena Sai** bearing Roll Number **164G1A0578**, **Y. Sreedhar Jaswanth** bearing Roll Number **164g1A05A1**, **S. Vijay Kumar** bearing Roll Number **164G1A05B7** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science & Engineering** during the academic year 2019-2020.

Signature of the Guide

Mr.M. Narasimhulu, M.Tech, (PhD)
Assistant Proffessor

Head of the Department

Dr.G.K. Venkata Narasimha Reddy,Ph.D
Professor & HOD

Date:

Place: Rotarypuram

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of people who made it possible, whose constant guidance and encouragement crowned my effort with success. It is a pleasant aspect that I have now the opportunity to express my gratitude for all of them.

It is with immense pleasure that I would like to express my indebted gratitude to my Guide **Mr.M. Narasimhulu**, M.Tech, (PhD), **Assistant Professor, Computer Science and Engineering**, who has guided me a lot and encouraged me in every step of the seminar work. I thank him for the stimulating guidance, constant encouragement and constructive criticism which have made possible to bring out this project work.

We express our deep felt gratitude to **Mr.M. Narasimhulu**, M.Tech, (PhD), **Assistant Professor**, project coordinator valuable guidance and unstinting encouragement enabled us to accomplish our project successfully in time.

We are very much thankful to **Dr. G. K. Venkata Narasimha Reddy**, PhD, **Professor and Head of Department, Computer Science & Engineering**, for his kind support and for providing necessary facilities to carry out the work.

We wish to convey my special thanks to **Dr. T. Hitendra Sarma**, Ph.D, **Principal, Srinivasa Ramanujan Institute of Technology** for giving the required information in doing our project work. Not to forget, we thank all other faculty and non-teaching staff, and my friends who had directly or indirectly helped and supported me in completing our project in time.

We also express our sincere thanks to the Management for providing excellent facilities.

Finally, we wish to convey our gratitude to our family who fostered all the requirements and facilities that we need.

(N. Pragathi)
(K. Praneetha)
(B. Raveena Sai)
(Y. Sreedhar
Jaswanth)
(S.Vijay Kumar)

DECLARATION

We, Ms N.Pragathi with reg no: 164G1A0568, Ms K. Praneetha with reg no: 164G1A0569, Ms B.Raveena Sai with reg no:164G1A0578, Mr Y. Sreedhar Jaswanth with reg no:164G1a05A1, Mr S.Vijay Kumar with reg no:164G1a05B7 students of SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY, Rotarypuram, hereby declare that the dissertation entitled “DRIVER’S DROWSINESS SYSTEM” embodies the report of our project work carried out by us during IV year Bachelor of Technology in Mr. M. Narasimhulu M.Tech,(PhD), Department of CSE, SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY, and this work has been submitted for the partial fulfillment of the requirements for the Bachelor of Technology degree.

The result embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

N.PRAGATHI

Reg no: 164G1A0568

K.PRANEETHA

Reg no: 164G1A0569

B.RAVEENA SAI

Reg no: 164G1A0578

Y.SREEDHAR JASWANTH

Reg no: 164G1A05A1

S.VIJAY KUMAR

Reg no: 164G1A05B7

CONTENT

	Page no
Abstract	vi
List of Figures	vii
List of Screens	xi
1. Introduction	1
1.1 Motivation	1
1.2 Problem definition	1
1.3 Objective of project	1
1.4 Limitations of project	2
1.5 Organization of documentation	2
2. Literature survey	3
2.1 Introduction	3
2.2 Existing System	3
2.3 Proposed System	5
2.4 Conclusion	5
3. Analysis	6
3.1 Introduction	6
3.2 Software Requirement Specification	6
3.2.1 User Requirements	8
3.2.2 Software requirements	8
3.3 Installation of python	8
3.4 Installation of OpenCV	13
3.5 Hardware requirements	17
4. Design	19
4.1 Introduction	19
4.2 Proposed Architecture	20
4.3 Description of modules	21
4.4 UML Diagrams	21
4.4.1 Usecase Diagram	23
4.4.2 Sequence Diagram	24
4.4.3 Collaboration Diagram	25
4.5 Data Flow Diagram	26
5. Implementation	29
5.1 Introduction	29
5.2 Workflow of the proposed system	29
5.2.1 Loading the facial landmark predictor and EAR	30
5.2.2 Capturing the frames	30
5.2.3 BGR to gray conversion	31
5.2.4 Calculating the Euclidean distance	32
5.2.5 Construction of contours	32
5.2.6 Detecting blinks	33
5.2.7 Drowsiness Detection	34
5.3 coding	35

6.	Testing and Validation	48
	6.1 Introduction	48
	6.2 Principles of testing	48
	6.3 Types of software testing	49
	6.3.1 Black Box Testing	49
	6.3.2 White Box Testing	50
	6.4 Test cases	51
7.	Results	53
	Conclusion	53
	Bibliography	54

ABSTRACT

A Driver Drowsiness Detection System is a real time problem solving system, which is capable of detecting the drowsiness or the fatigue of the driver while driving. The fatigue for the driver is caused due to the continues long drives without any rest which apparently give rise to the accidents. According to a survey held, the maximum number of accidents have been taken place due to fatigue of the driver while driving to the long distances without any break. Here, in this paper we propose a system which ensures the safety of the diver by monitoring the status of the driver's eye and alerting the driver whenever necessary without fail. Also, we have two stages of alerting the driver, in the first stage of alert, we alert by focusing on the eye blink rate of driver where as in the second stage of alert, we go through a process which monitors the status of driver's eye indicating that the driver is drowsy when he falls asleep.

